Review retrospective: *the generic-(k_1, k_2) partition lift, and a doc bullet that named the wrong theorem*

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Abstract

A soundness review of `Laplace/TwoD/KthKthPartitionAsymptotic.lean`: the generic-(k_1, k_2) asymptotic of the 2D separable monomial partition function, the first leg of the "2D lift" trio built on the just-reviewed AddSeparable root. Result: **a sound formalisation with one dead-code removal and one docstring-accuracy fix** — a dead `open Real`, and a module bullet that mislabelled which theorem produces the closed form (and called a fractional power-law a "polynomial").

1 Setting

This file instantiates the four-step "2D lift" pattern documented in the seabed `CLAUDE.md`: factorise the 2D quantity into a product of two 1D ones (via AddSeparable), substitute the 1D closed form, peel the t-dependence, and package as a rescaled `Tendsto` and an `IsEquivalent`. It lifts the fixed $(k_1, k_2) = (2, 3)$ quartic–sextic partition file to parametric (k_1, k_2). Reviewing it right after its factorisation root made the fidelity check fast — the only genuinely new content is the composite constant and the exponent arithmetic.

2 What was reviewed

Five public declarations: the composite constant $C(k_1, k_2) = \frac{1}{k_1 k_2}(2k_1)!^{1/(2k_1)}(2k_2)!^{1/(2k_2)}\Gamma(\frac{1}{2k_1})\Gamma(\frac{1}{2k_2})$, and the four packaged theorems — the exact factorisation $Z_{2D} = Z_{L_{k_1}} Z_{L_{k_2}}$, the $\texttt{const}\cdot t^{-1/(2k_1)-1/(2k_2)}$ form, the rescaled limit $\to C$, and the asymptotic equivalence. The corpus was the file docstrings, the `CLAUDE.md` architecture section, the factorisation root, the 1D base (MonomialPotential), and the $(2, 3)$ sibling it lifts.

3 Fidelity / soundness findings

The mathematics is sound. The composite constant is exactly the product of the two 1D `partitionFunction_kthPotential` constants; the exponent $-1/(2k_1) - 1/(2k_2)$ is correct in both the exact and equivalence statements (and its positive mirror in the rescaling); the hypotheses sit where they should ($t > 0$ for the exact equalities, only $k_i \geq 1$ for the `atTop` statements).

One docstring inaccuracy, fixed. The module "Headline results" bullet for `partitionFunction_kthKth_eq` read "exact closed form $Z_{2D} = Z_{L_{k_1}} Z_{L_{k_2}}$ *specialised to a scalar* *polynomial-in-t form.*" Two errors: (i) that theorem proves *only* the factorisation — the scalar `const`$\cdot t^{(\cdots)}$ form is the *next* theorem (`partitionFunction_kthKth_eq_const_mul_rpow`), which the bullet beneath it already describes; (ii) the closed form is a *fractional* power-law $t^{-1/(2k_1)-1/(2k_2)}$, not a polynomial. Corrected to "exact factorisation … into the two 1D partitions." Comment-only, signatures byte-identical.

4 Simplifications applied

One dead `open`. Every `Real.exp`/`Real.div_rpow`/`Real.rpow_*`/`Real.Gamma` is qualified, and the bare t^(real) is the `HPow`/`rpow` instance, which does not require `open Real`; the other four opens (`MeasureTheory` for bare `Integrable`, `Filter` for `Tendsto`/`atTop`, `Asymptotics` for $\sim$ [atTop], scoped `Topology` for $\mathcal{N}$) are load-bearing. Removal: module green at 2700 jobs, no warnings. GPT affirmed.

5 Disagreements and open questions

None. The reviewer's remaining suggestions (`simpa`-shorten the $n = 0$ integrability sub-proofs; replace the inline `show ... by ring` equality-proofs inside `rw` with `change`) were declined as proof-style refactors: notably, `change` cannot replace an equality *proof term* passed to `rw`, and a term-mode `show` inside `rw` is idiomatic and not what `linter.style.show` flags, so that one is not even warning-clearing.

6 Lessons

- **A "headline results" bullet can name the wrong theorem.** When a file lists several closely-related theorems (factorisation, then the closed form, then the limit), a summary bullet can attach one theorem's description to its neighbour. Check each bullet against the declaration it names, not against the file's overall thrust — and watch loose words like "polynomial" for a quantity that is actually a fractional power.
- **Reviewing a lift right after its root is cheap and high-confidence.** With AddSeparable just reviewed, this lift's only new surface was the constant and the exponent algebra; the factorisation step was already trusted. Dependency-ordered review keeps each fidelity check small.

7 Follow-ups

- The remaining legs of the lift trio (`KthKthNumeratorAsymptotic`, `KthKthMomentAsymptotic`, `KthKthOddMoments`) and the $(2, 3)$ `QuarticSextic*` specialisations are now reviewable with the same template — and a near-certain candidate for the same dead `open Real` (per the Quartic→Sextic prediction pattern).
- Proof-golf forward tide: the `simpa` integrability shortenings the reviewer flagged.